# Ashish Shajan

**Lead AI/ML Engineer, AT Bots**

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## Profile

Lead AI/ML Engineer at AT Bots with hands-on work in deep learning, computer vision and edge AI for autonomous robots and drones. Trained a crack-detection model from scratch and integrated it on an NVIDIA Jetson companion computer of an autonomous quadcopter that completed real-hardware flight tests and a field trial. Builds complete AI pipelines: data collection and annotation, preprocessing, model training and evaluation, onboard inference and integration with ROS 2 and PX4. Also experienced in classical machine learning, LLMs and prompt engineering.

## Professional Experience

Lead AI/ML Engineer and Chief Technical Officer (Apr 2025 – Present); AI/ML Engineer (Jan – Apr 2025)
*(AT Bots, Kerala)* *January 2025 – Present*

- **Autonomous structural crack-detection drone:** trained the crack-detection deep learning model from scratch and integrated it with the camera pipeline on an NVIDIA Jetson companion computer; led system integration of the quadcopter, PX4 flight controller, ROS 2 and the ground station.
- Completed autonomous flight tests on real drone hardware and a field trial of the inspection system.
- Worked on sensor integration and the sensor-fusion design for drone localization; ran simulation tests (Gazebo, PyBullet) before hardware tests.
- **EduBot:** technical integration of AT Bots' AI-powered education and robotics product (app, software and robot hardware) for STEM learning.
- Leads the AT Bots AI/ML engineering team: model development, data pipelines, code review and hardware/software integration.

Captain and Autonomous Head
*(Team Equinox, SAE aBAJA)* *2025*

- Led the Level 2 autonomous BAJA vehicle: ROS 2 stack for perception, sensor fusion (Radar, IMU, wheel encoders), EKF localization and ADAS control (ACC, AEB, LKA); validated in Gazebo, CARLA and IPG CarMaker.

Research Intern, Hyderabad, India
*(Robotics Research Centre, IIIT Hyderabad)* *Oct 2024 – Aug 2025*

- Built Gazebo simulation environments for autonomous drone control; built an autonomous mobile robot with 2D LiDAR SLAM, wheel odometry, HITL testing and PID tuning.

Intern - INTEL Unnati Industrial Training Program, Bengaluru, India
*(INTEL Corporation Bangalore)* *May 2024 – Jul 2024*

- National-level industrial training in AI and data-driven systems: knowledge representation, data preprocessing, feature engineering, model training and validation (see Heart Attack Risk Prediction).

## AI/ML Projects

Autonomous Structural Crack-Detection Drone AT Bots
*(Custom deep learning model, NVIDIA Jetson, ROS 2, PX4, QGroundControl, quadcopter)*

- Quadcopter that flies an inspection mission, captures images of structural surfaces and detects cracks onboard with a custom model trained from scratch, to reduce manual visual inspection.
- Pipeline: dataset collection and annotation, preprocessing and augmentation, training and evaluation, export to the Jetson, onboard inference and logging of detections. Flight stabilisation stays on the PX4 flight controller; the companion computer runs perception.
- Status: real-hardware autonomous flights and a field trial completed. Next step: defect localization and mapping.

Heart Attack Risk Prediction using Machine Learning, Intel Unnati Project
*(Python, Scikit-learn, Pandas, Logistic Regression, K-Means)* *2024*

- End-to-end ML pipeline on 303 patient records (13 clinical features): data cleaning, z-score outlier detection, EDA, knowledge graphs, K-Means clustering, classification and regression.
- Logistic Regression: 86.8% accuracy, F1 0.88, ROC-AUC 0.87. Code: github.com/Ashish480/Heartattackprediction

EduBot: AI-Powered Education and Robotics Platform AT Bots Product
*(Educational app and software, robot hardware, embedded systems)*

- Education-focused robotics product that combines learning software with a robot for practical STEM education; contributed to technical integration of the app, software and hardware.

AI-Powered Waste Segregation Competition

*(Machine Learning, Computer Vision)* *2025*

- ML-based waste classification for automatic segregation. 3rd place, Vruthi 2025 Clean Kerala Conclave Wasteathon (Rs. 25,000).

Autonomous Perception Stack, BAJA SAE India Competition

*(ROS 2, Radar perception, EKF sensor fusion)* *2025*

- Radar-based perception and multi-sensor fusion for real-time localization; ADAS functions on ROS 2. Code: github.com/Ashish480/Carla_abaja

**Robotics and simulation:** drone simulation and control in PyBullet (drone spawning, teleoperation, autonomous circular flight, IMU feedback and PID control); 12-DOF quadruped robot (ROS 2, Isaac Sim); differential drive robot with Cartographer SLAM; manipulator and self-balancing robot; underwater surveillance system.

## AI/ML Skills

- **Deep learning and model training:** Design and train neural network models from scratch; dataset splits, data augmentation, hyperparameter tuning, evaluation with precision, recall, F1 and IoU; PyTorch, YOLOv8, U-Net++, EfficientNet.
- **Computer vision:** Camera-based perception, image acquisition and preprocessing, crack and defect detection; classification, object detection and segmentation methods; detection visualization.
- **Edge AI deployment:** Model export and onboard inference on NVIDIA Jetson; latency and accuracy trade-offs; integration of AI output with ROS 2 nodes and mission software.
- **Data for AI:** Data collection, cleaning, annotation and quality checks; feature engineering; exploratory data analysis (Pandas, NumPy, Matplotlib, Seaborn).
- **Classical machine learning:** Logistic regression, decision trees, K-Means clustering, linear regression, model validation (Scikit-learn).
- **LLMs:** Large language models, prompt engineering and LLM-based application concepts.

## Robotics & Software Skills

- **Autonomy:** ROS 2, PX4 Autopilot, MAVROS / MAVLink, QGroundControl, Nav2, SLAM (LiDAR), localization, sensor fusion (IMU, GNSS, LiDAR, Radar, EKF), mission planning
- **Simulation & control:** Gazebo, NVIDIA Isaac Sim, PyBullet, CARLA, IPG CarMaker, RViz, MATLAB; PID control, trajectory generation, state estimation
- **Embedded:** NVIDIA Jetson, ESP32, Raspberry Pi, Arduino, Teensy, micro-ROS, sensor interfacing
- **Programming & tools:** Python, C, C++, Embedded C, Linux, Git, Docker, PlatformIO

## Education

**B.Tech in Robotics and Automation,** Saintgits College of Engineering, Kottayam. CGPA 7.6 2026

Courses: AI, Machine Learning, Computer Vision, Control Systems, Robot Kinematics & Dynamics, Embedded Systems, Mobile Robots, ROS. Higher Secondary: Jawahar Navodaya Vidyalaya, Wayanad.

## Certifications & Achievements

Intel Unnati Industrial Training (2024) • 3rd Place, Vruthi 2025 Wasteathon: AI waste segregation (Rs. 25,000) • 3rd Place, National Hackathon, Rajagiri School of Engineering • NPTEL: Soft Skill Development • Languages: English (professional), Malayalam (native), Hindi (intermediate)